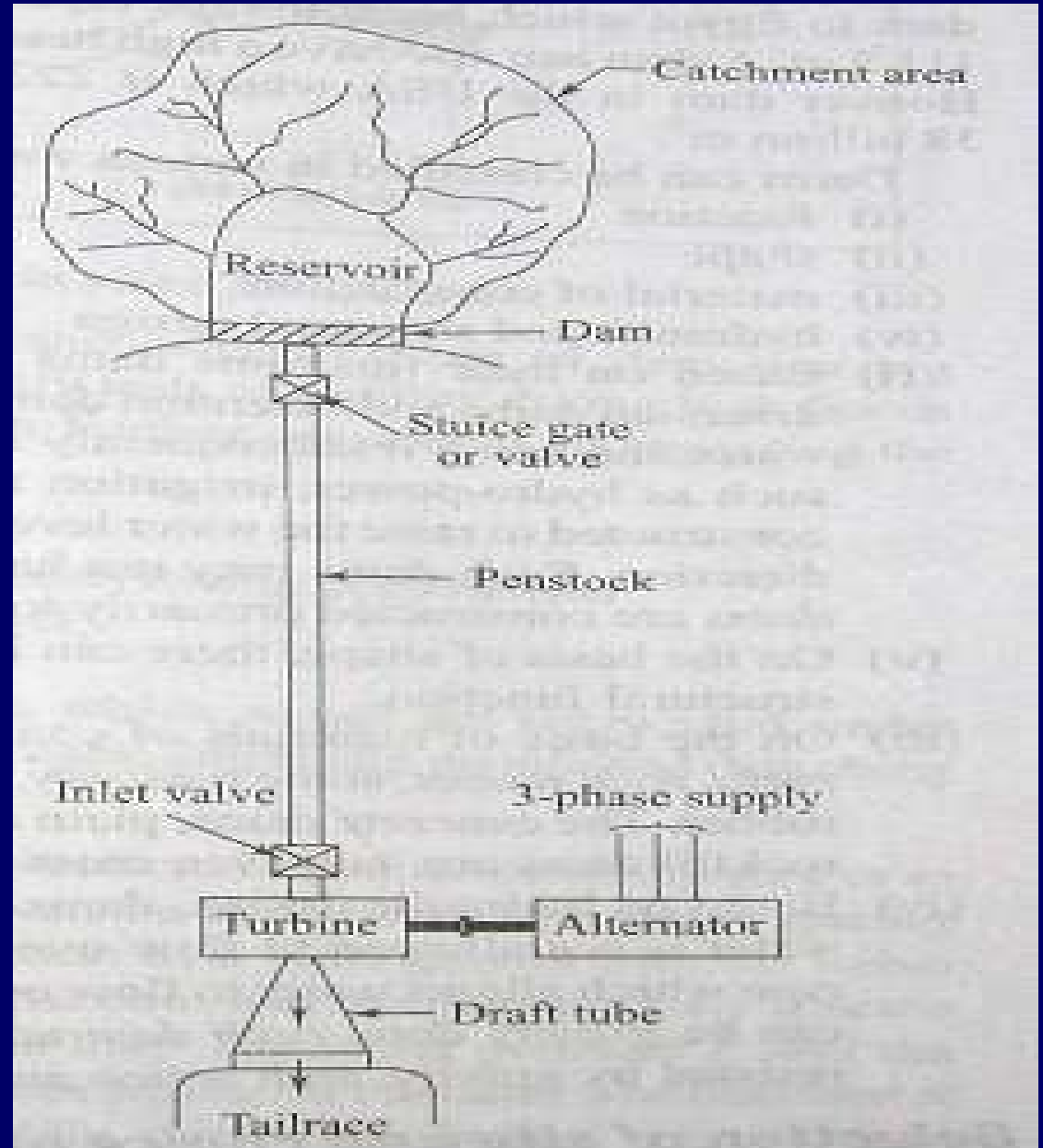


Power Plant Engg-Module 2

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CUSAT**

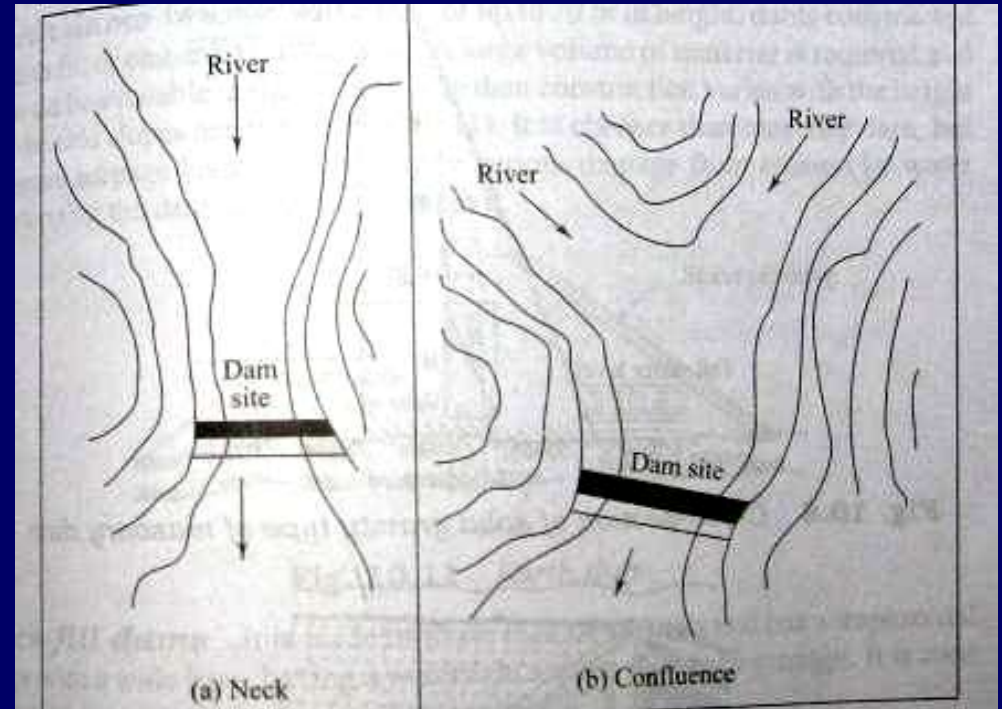
Hydro-electric power station

- Catchment area
- Reservoir
- Dam
- Spillways
- Conduits
- Surge tanks
- Penstock
- Draft tubes
- Power house



Dams

- Store water
- Builds up head
- Classification
 - Function
 - Shape
 - Material of construction
 - Hydraulic and structural design
- Selection of site and choice of construction



Suitable sites for a dam

Classification based on Function

- Surface Dam
- Underground dam
- Saddle Dam
- Weir
- Check Dam
- Dry dam

Surface Dam

- Structures over ground
- For Power Generation ,irrigation Flood Control etc

Under Ground dam

- Underground dams are used to trap groundwater and store all or most of it below the surface for extended use in a localized area. In some cases they are also built to prevent saltwater from intruding into a freshwater aquifer. Underground dams are typically constructed in areas where water resources are minimal and need to be efficiently stored, such as in deserts and on islands

Fukuzato Dam Okinawa



Saddle dam

- A saddle dam is an auxiliary dam constructed to confine the reservoir created by a primary dam either to permit a higher water elevation and storage or to limit the extent of a reservoir for increased efficiency. An auxiliary dam is constructed in a low spot or saddle through which the reservoir would otherwise escape

Saddle Dam



Weir

- A weir is a barrier across a river designed to alter the flow characteristics. In most cases, weirs take the form of a barrier, smaller than most conventional dams, across a river that causes water to pool behind the structure (not unlike a dam) and allows water to flow over the top



Check dam

- A check dam is a small dam, which can be either temporary or permanent, built across a minor channel or drainage ditch. Similar to drop structures in purpose, they reduce erosion and gullyng in the channel and allow sediments and pollutants to settle

Dry dam

- A dry dam also known as a flood retarding structure, is a dam designed to control flooding. It normally holds back no water and allows the channel to flow freely, except during periods of intense flow that would otherwise cause flooding downstream.

Types of Dams Based on Hydraulic Design

- Solid Gravity Dam
- Hollow Gravity Dam
- Buttress Dam
- Arch Dam
- Embankment Dam
- Earth Dam

Solid Gravity Dam

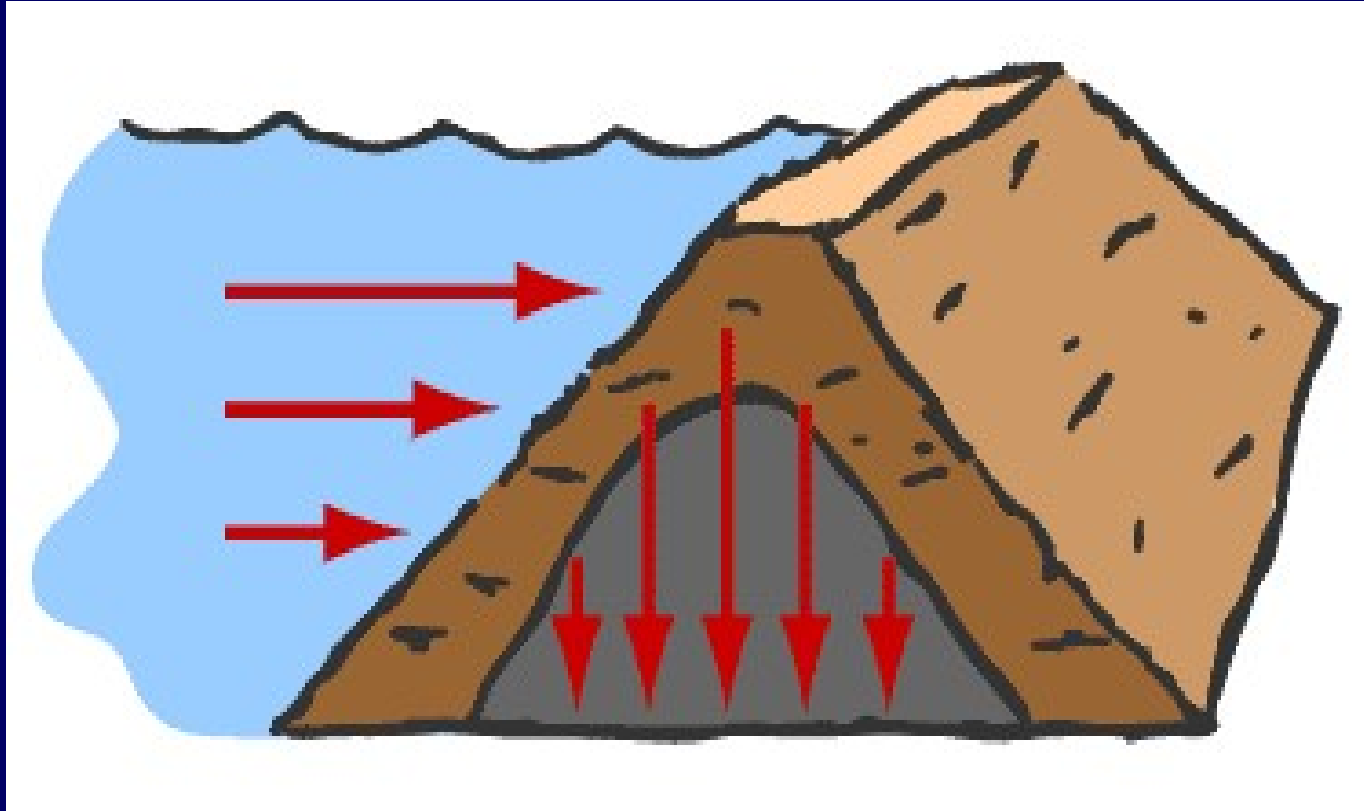




Buttress dam



Embankment Dam



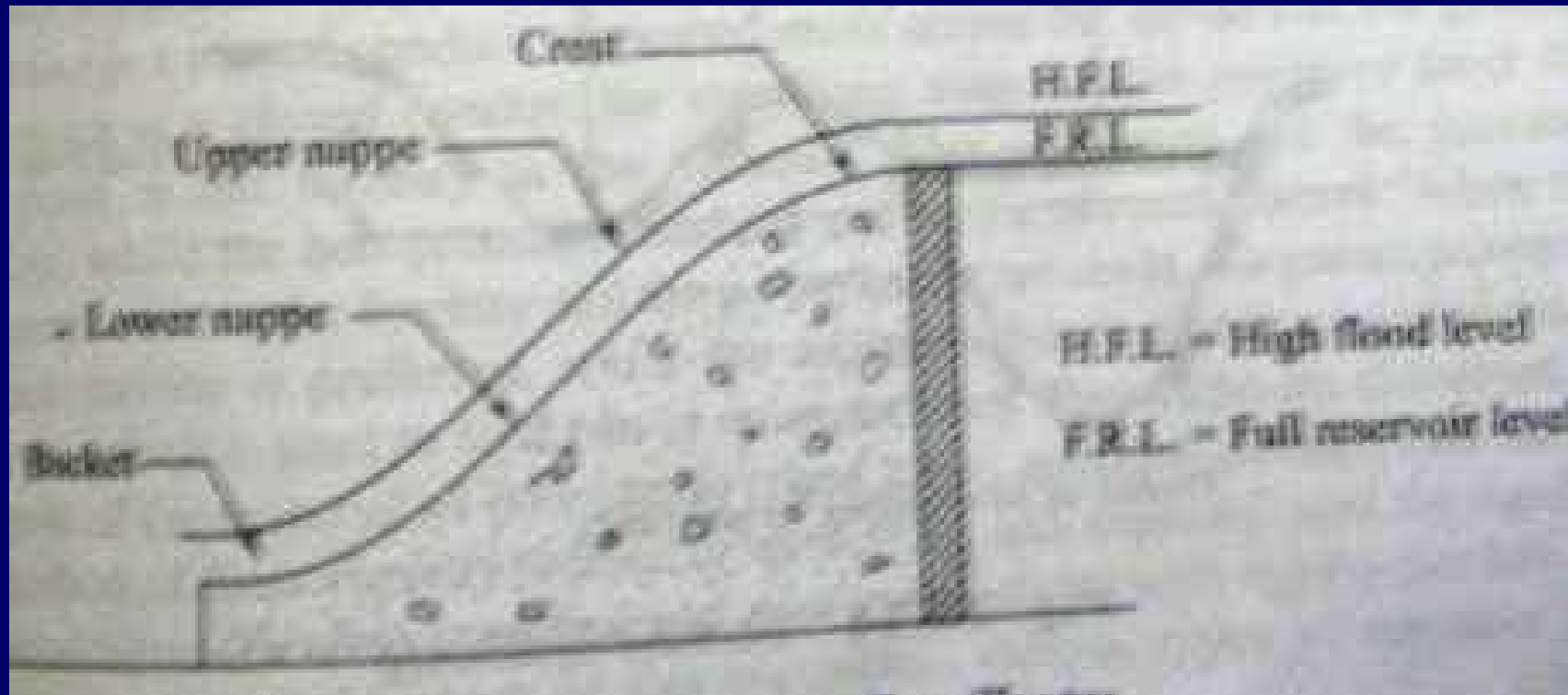
Earth dam



Spillway

- A spillway is a structure used to provide the controlled release of flows from a dam or levee into a downstream area, typically being the river that was dammed

Spill way



- Remove excess water
- Provides structural safety

Chute Spill way



Stepped Spill way



Bell mouth spill way



PENSTOCKS-1

- A penstock is a sluice or gate or intake structure that controls water flow, or an enclosed pipe that delivers water to hydraulic turbines and sewerage systems. It is a term that has been inherited from the technology of wooden watermills

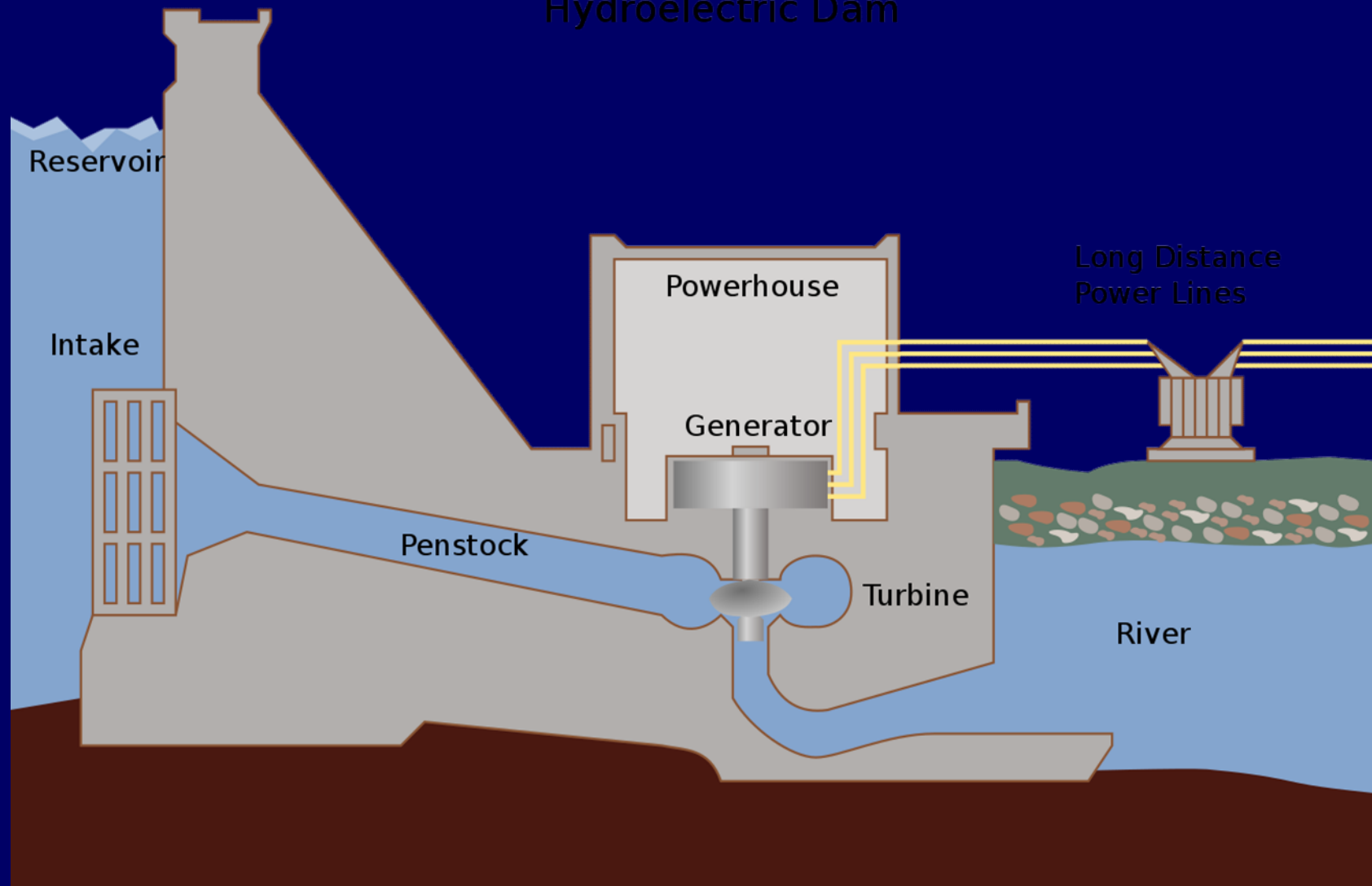


PENSTOCKS-2

- **Penstocks** are pipes or long channels that carry water down from the hydroelectric reservoir to the turbines inside the actual power station.^[2] Generally, they are made of steel and water under high pressure flows through the penstock. They are a vital component of a hydroelectric facility that allows water to move to the turbine. Grates or filters can be attached to the ends of penstocks to trap large debris such as branches. This ensures that debris cannot enter the channel and block it.
- The amount of water that is allowed to flow through the penstock can be controlled with a **sluice**, which is simply a gate that can be raised and lowered to increase or decrease the amount of water allowed to flow through. When the sluice is fully open, water flows freely down through the penstock. However, when it is closed slightly there is a limitation to how much water can flow, and thus less water enters the penstock

PENSTOCK 3

Hydroelectric Dam



PENSTOCK-4



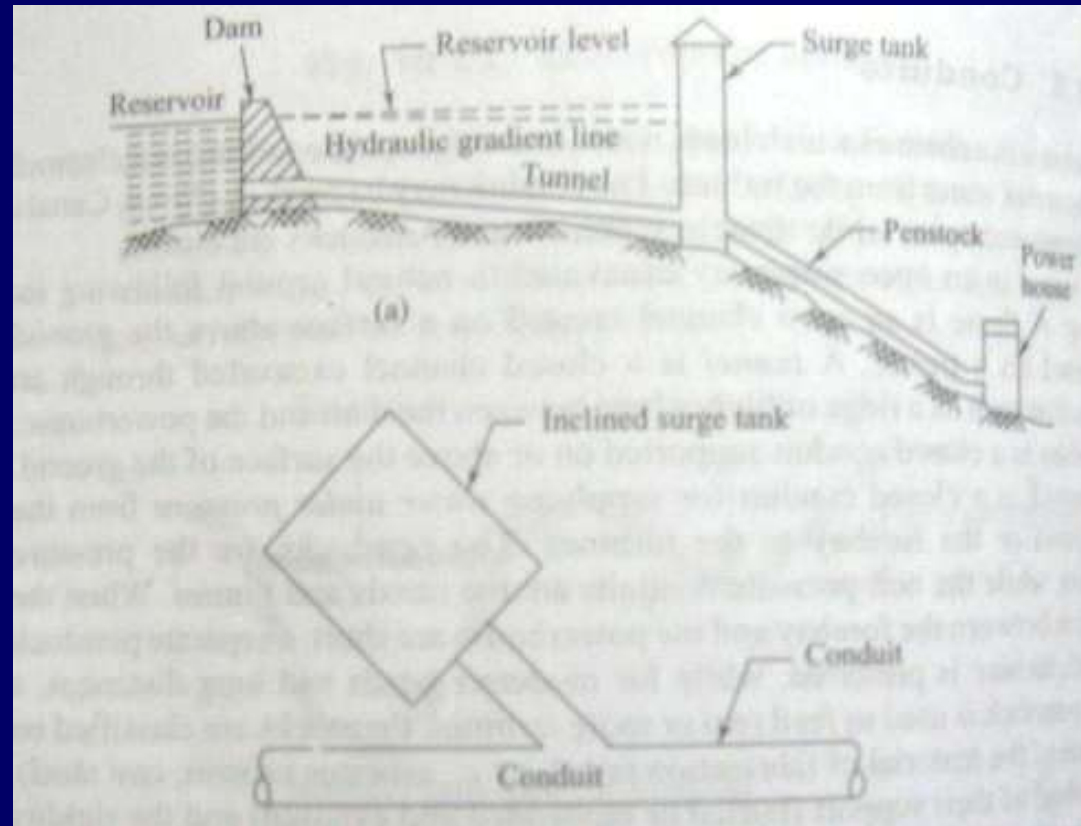
PENSTOCK-5



SURGE TANKS

- A **Surge tank** is a water storage device used as a pressure neutralizer in hydropower water conveyance systems in order to dampen excess pressure variance.
- A surge **tank** (or **surge drum** or **surge pool**) is a standpipe or storage reservoir at the downstream end of a closed aqueduct, feeder, dam, barrage pipe to absorb sudden rises of pressure, as well as to quickly provide extra water during a brief drop in pressure.
- Surge tanks are located near to the power house to reduce length of penstocks.
- No limitations regarding surge tank height.
- Location at which flat sloped conduit and steep sloped penstock meet

Surge tank



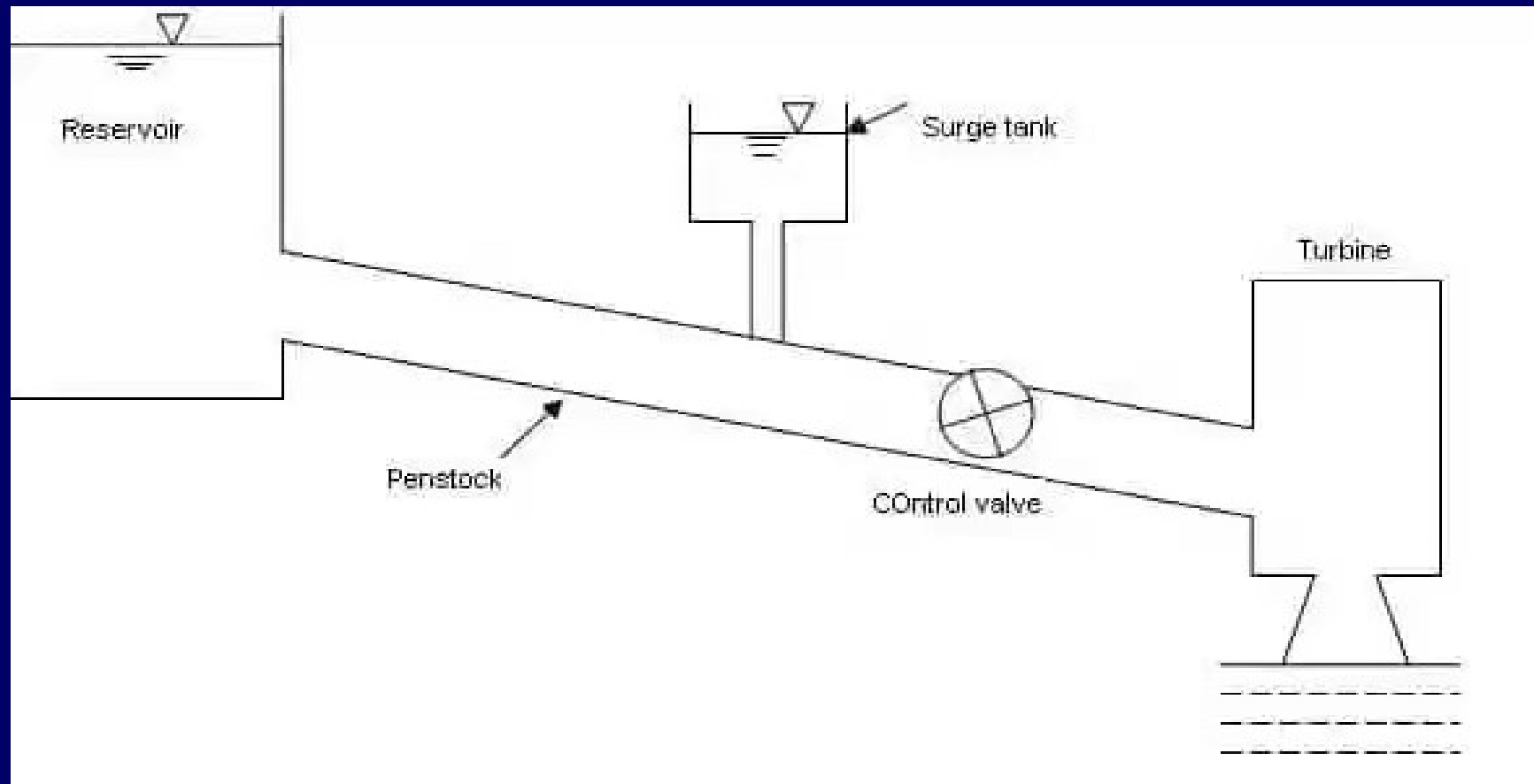
- Small reservoir
- Reduce pressure swing
- Needed for high head plants

Surge Tank

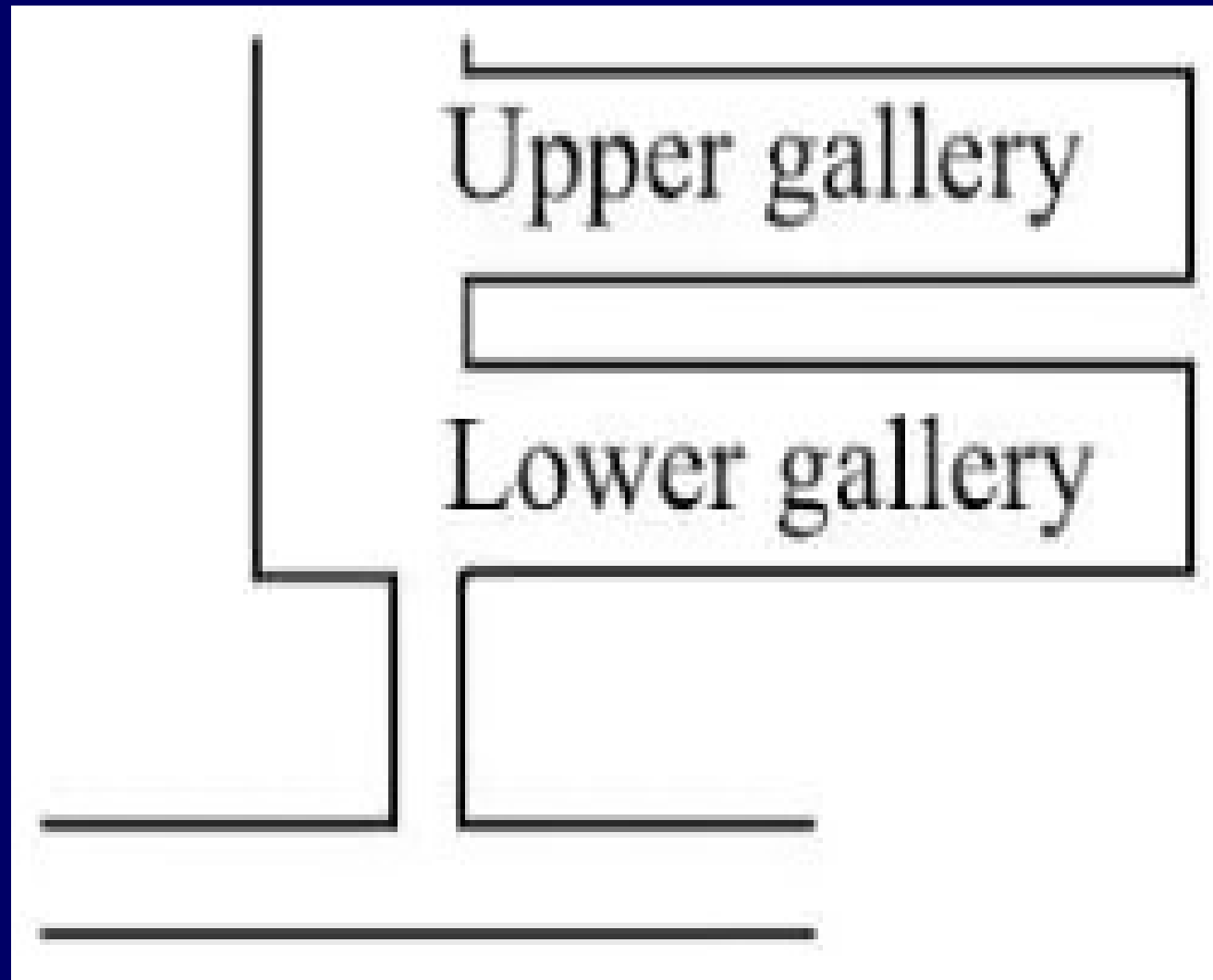


Types of Surge Tanks

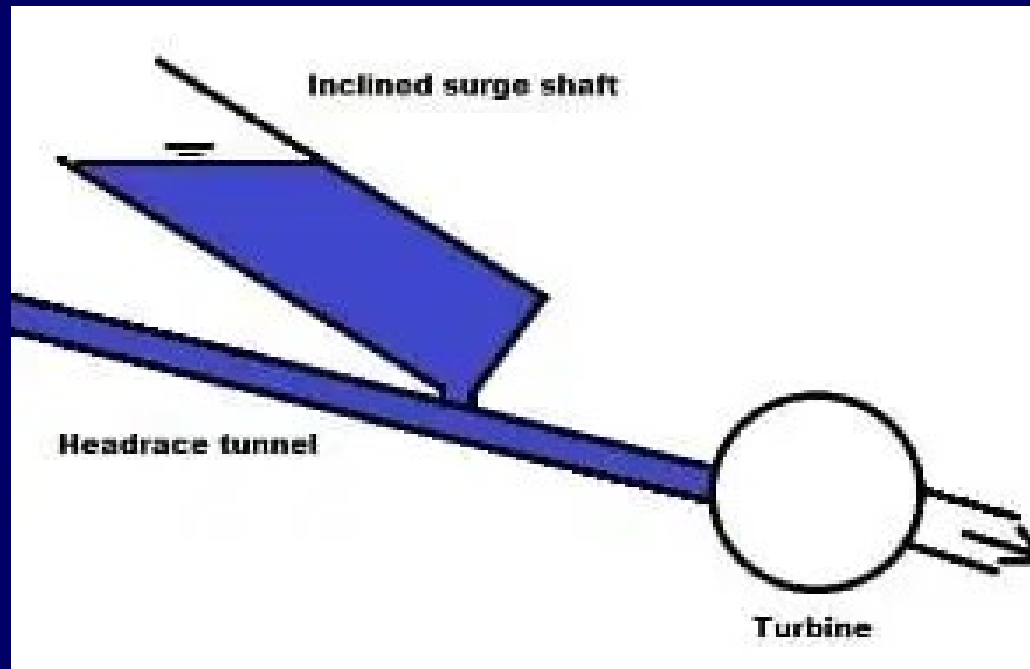
- Simple surge tank



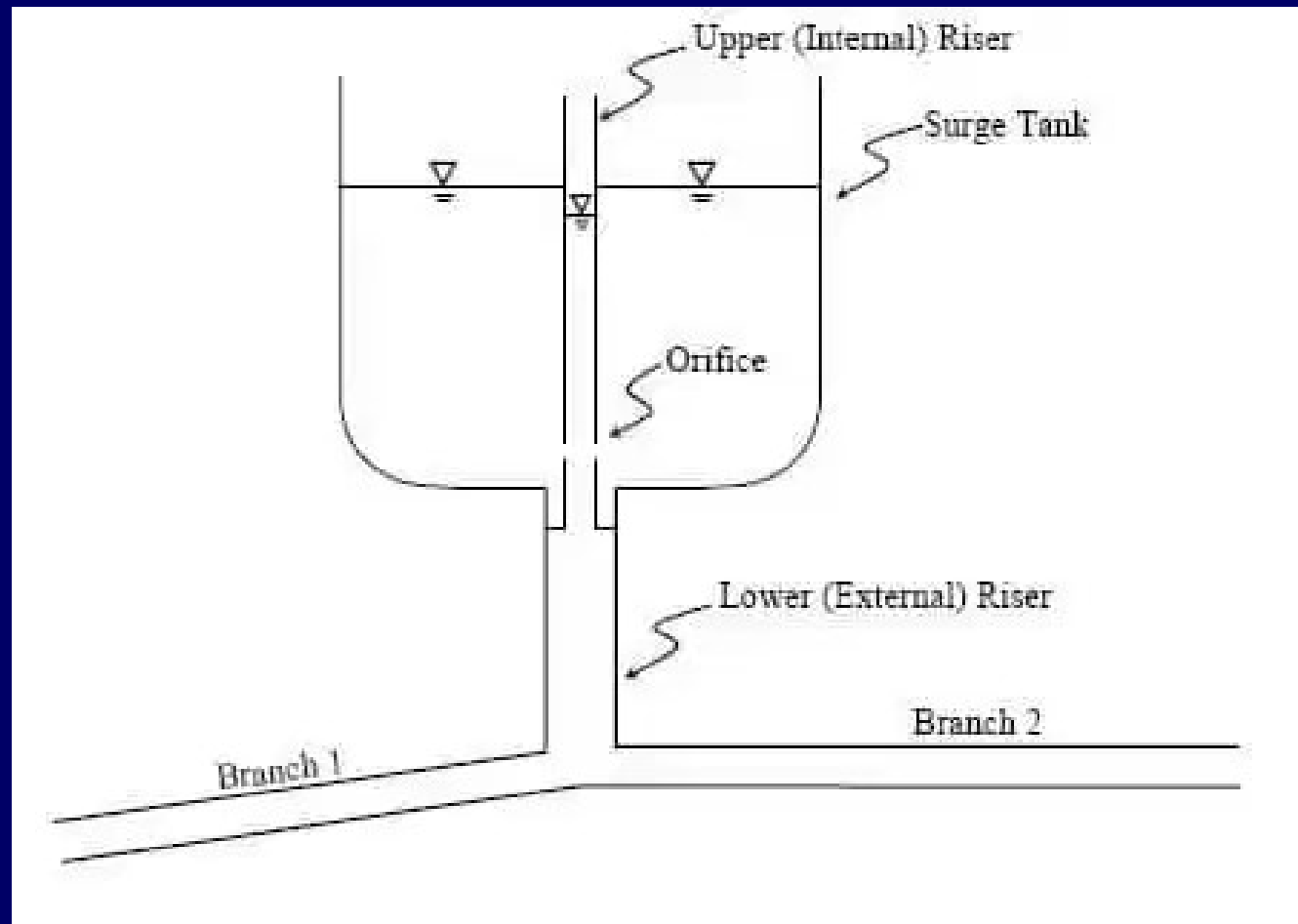
Gallery type surge tank



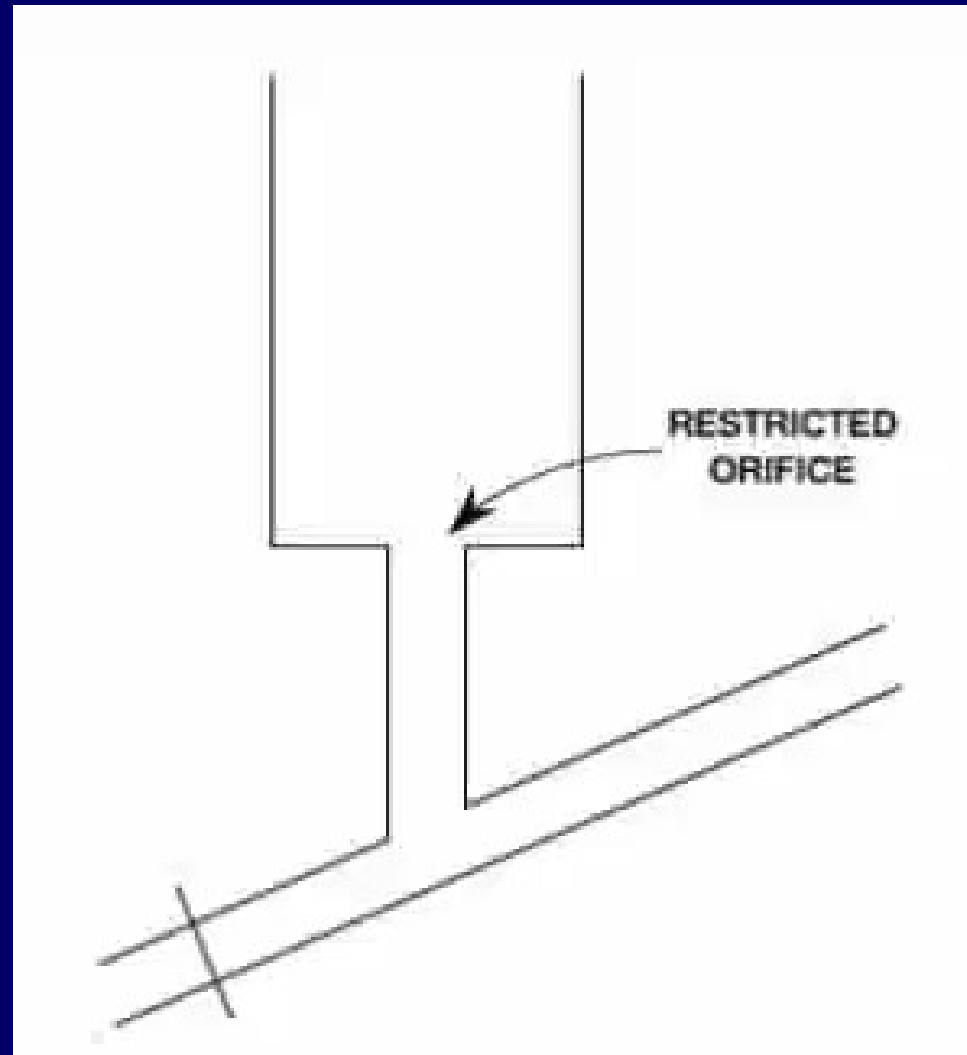
INCLINED TYPE SURGE TANK



DIFFERENTIAL SURGE TANK



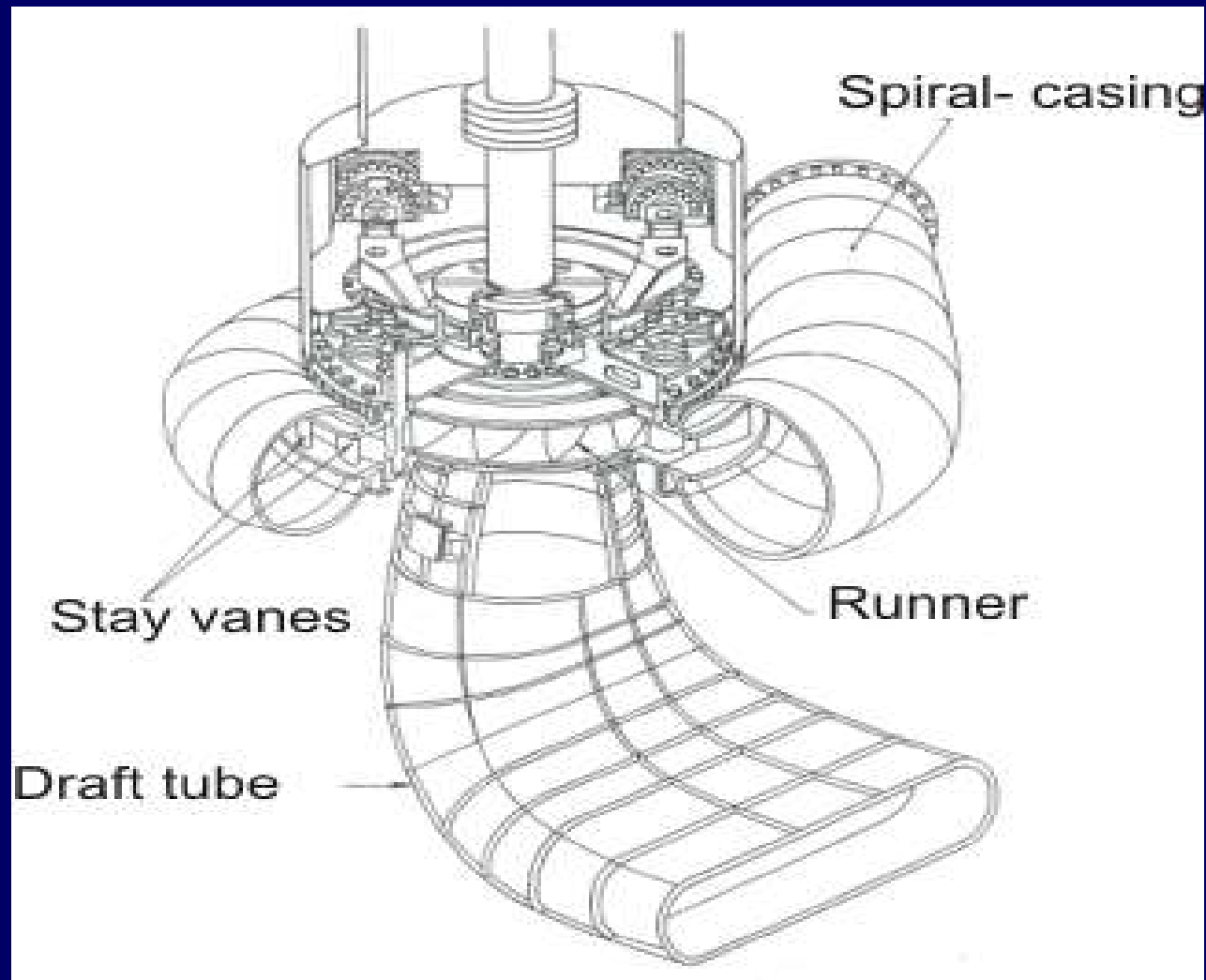
RESTRICTED ORIFICE SURGE TANK



Draft Tube

- The draft tube is a conduit which connects the runner exit to the tail race where the water is being finally discharged from the turbine. The primary function of the draft tube is to reduce the velocity of the discharged water to minimize the loss of kinetic energy at the outlet. This permits the turbine to be set above the tail water without any appreciable drop of available head

Draft Tube



DRAFT TUBE

- The principal purpose of the draft tube is to convert water kinetic energy into pressure energy. To decrease the velocity of the water and to raise the pressure of the water before joining the tailrace, the pipe is used to steadily increase the cross-sectional area. The draft tube raises the water pressure to the atmospheric pressure. The Draft Tube must be strong enough to withstand the high pressure and speed of the water. DT are made of cast steel and cemented concrete.

TYPES OF DRAFT TUBES

1. Conical draft tube
2. Simple elbow draft tube
3. Moody spreading draft tube
4. Elbow draft tube with a varying cross-section

Conical draft tube

- In this type of draft tube form, the flow direction is straight and divergent. This tube style is made of mild steel plates. It is tapered in shape and the outlet diameter is greater than the inlet diameter of the draft tube. The tapered angle of the draft tube should not be too wide to induce a divergence of the flow from the wall of the draft tube. This angle should also not be too short, since it would require a longer draft tube that brings a substantial loss of kinetic energy. So, the angle of the taper is still almost 10 degrees.

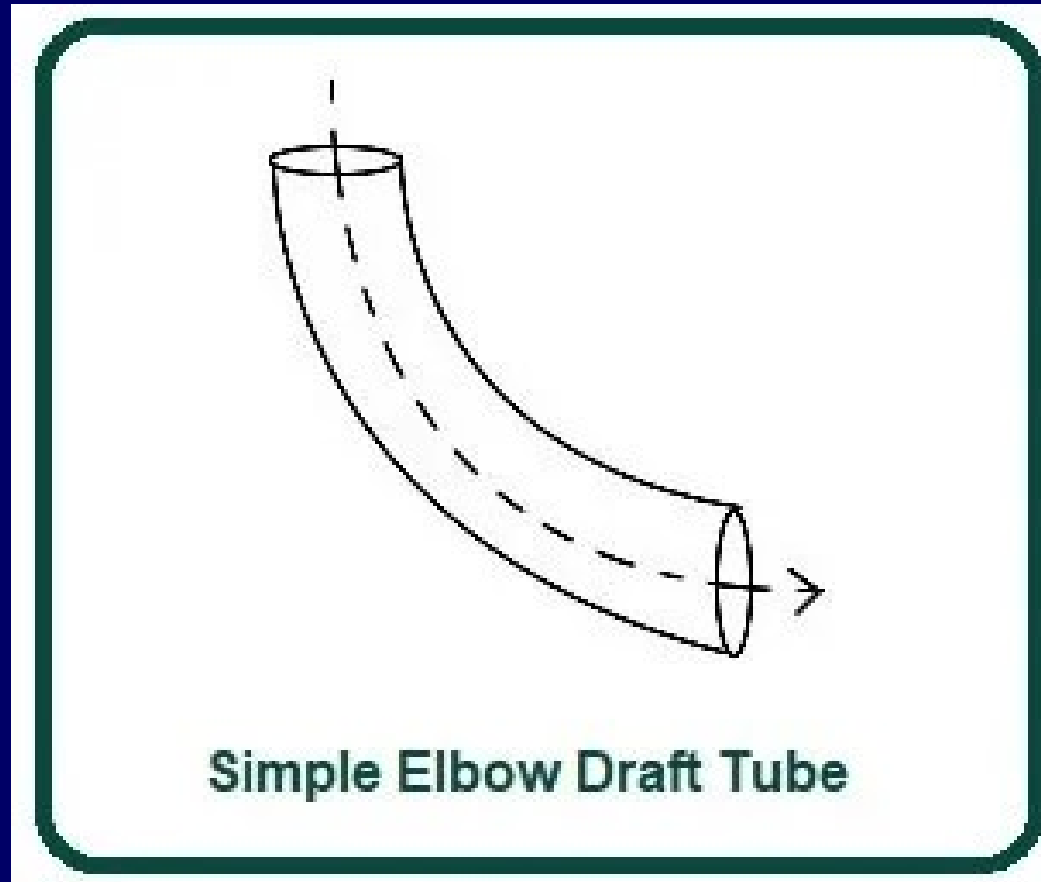
CONICAL TYPE DT



SIMPLE ELBOW DT

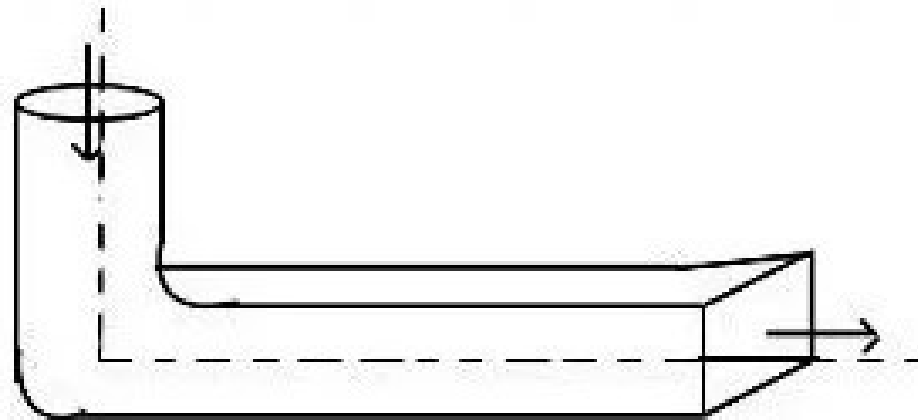
- In a simple elbow draft tube, the tube is elbow-shaped.
- It is mainly used in Kaplan turbines. In this type of draft tube, the cross-section area remains the same throughout the length of the draft tube.
- The inlet & outlet of the draft tube is circular. This draft tube is usually used at low head locations, and the turbine is to be placed close to the tailrace.
- This helps cut the cost of digging, and the exhaust diameter must be as large as possible to recover kinetic energy at the runner's outlet.

SIMPLE ELBOW TYPE DT



Elbow Draft Tube with Varying Cross Section

- The elbow draft tube with different cross-sections is an improvement of the simple elbow draft.
- In this type of draft tube, the inlets are circular, and the outlet is rectangular in shape.
- The horizontal portion of the draft tube is usually tilted upward to prevent air ingress from the exit end.
- The cross-section area of the type of draft tubes varies from inlet to outlet.
- The outlet of this draft tube is always below the tailrace.
- It is commonly used in Kaplan turbines, and the efficiency of this type of draft tube is about 70 percent.

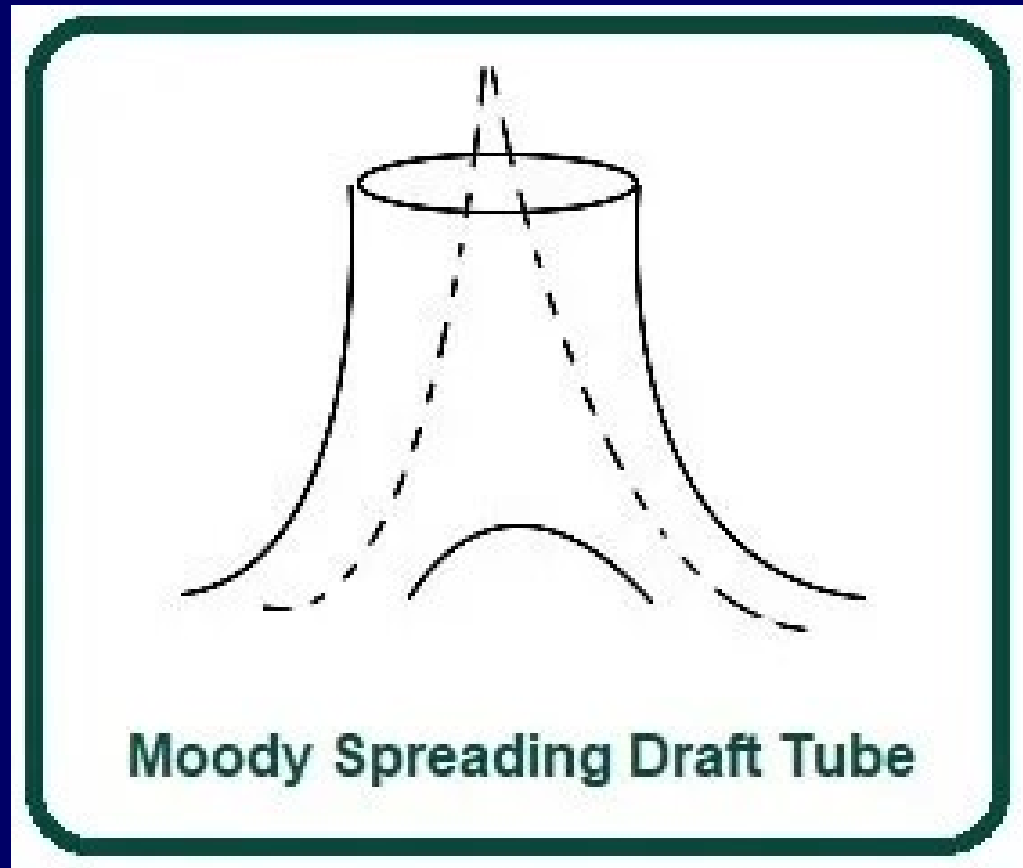


Elbow Draft Tube with Varying Cross Section

Moody Spreading Draft Tube

- In this type of draft tube, an outlet of the draft tube is divided into two parts.
- It is similar to a conical draft tube and is provided with a central core part that divides the outlet into two parts. This draft tube has one inlet and two outlets.
- This type of draft tubes is mainly used to reduce the swirling action of water.
- It is used in vertical shaft turbines. The efficiency of this type of draft tube is about 88 percent

MOODY DIFFUSING TYPE



Site Selection

- One of the best places for building a dam is a narrow part of a deep river valley; the valley sides can then act as natural walls. The primary function of the dam's structure is to fill the gap in the natural reservoir line left by the stream channel. The sites are usually those where the gap becomes a minimum for the required storage capacity.

Considerations of Site Selection

- permeability of the surrounding rock or soil
- earthquake faults
- landslides and slope stability
- water table
- peak flood flows
- reservoir silting
- environmental impacts on river fisheries, forests and wildlife
- impacts on human habitations
- compensation for land being flooded as well as population resettlement